Churn Analysis of TelCom

A Presentation by Royal Challengers BI

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Introduction

- Customer Churn Analysis
- Predictive model using Orange and Power-BI
- Dataset: 7,043 rows, 21 labels

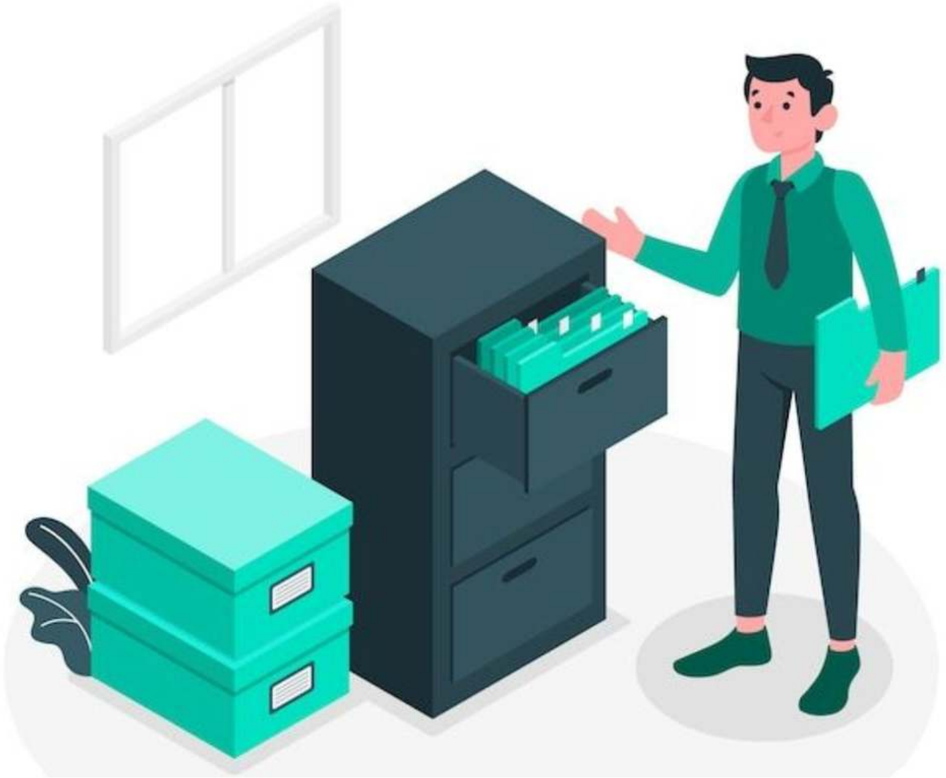
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Problem Statement

The objective of the team is to develop a predictive model using Orange and Power-BI. We plan on analyzing customer data and classify customers who are likely to churn.

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Use Case

- Addressing the customer churn
- Impact on revenue and profitability
- At-risk customers
- Whether the customer has left within the last month

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Data Overview and Preprocessing

Source: Kaggle – Telco Customer Churn

- 7,043 rows, 21 features.

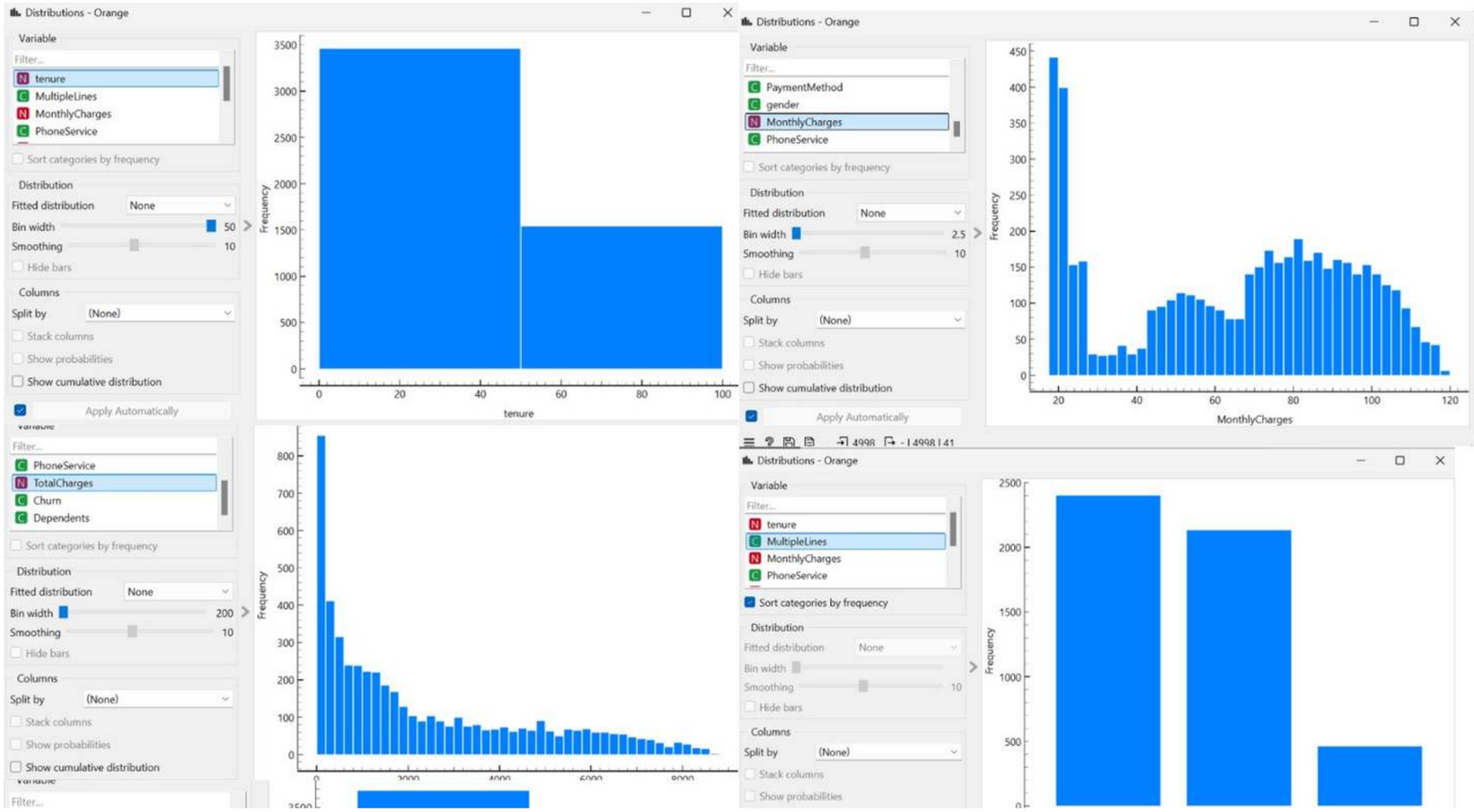
Key Variables:

- Churn – Whether the customer left
- Tenure – How long they stayed
- MonthlyCharges – Monthly payment
- Contract – Type of plan
- PaymentMethod – How they pay



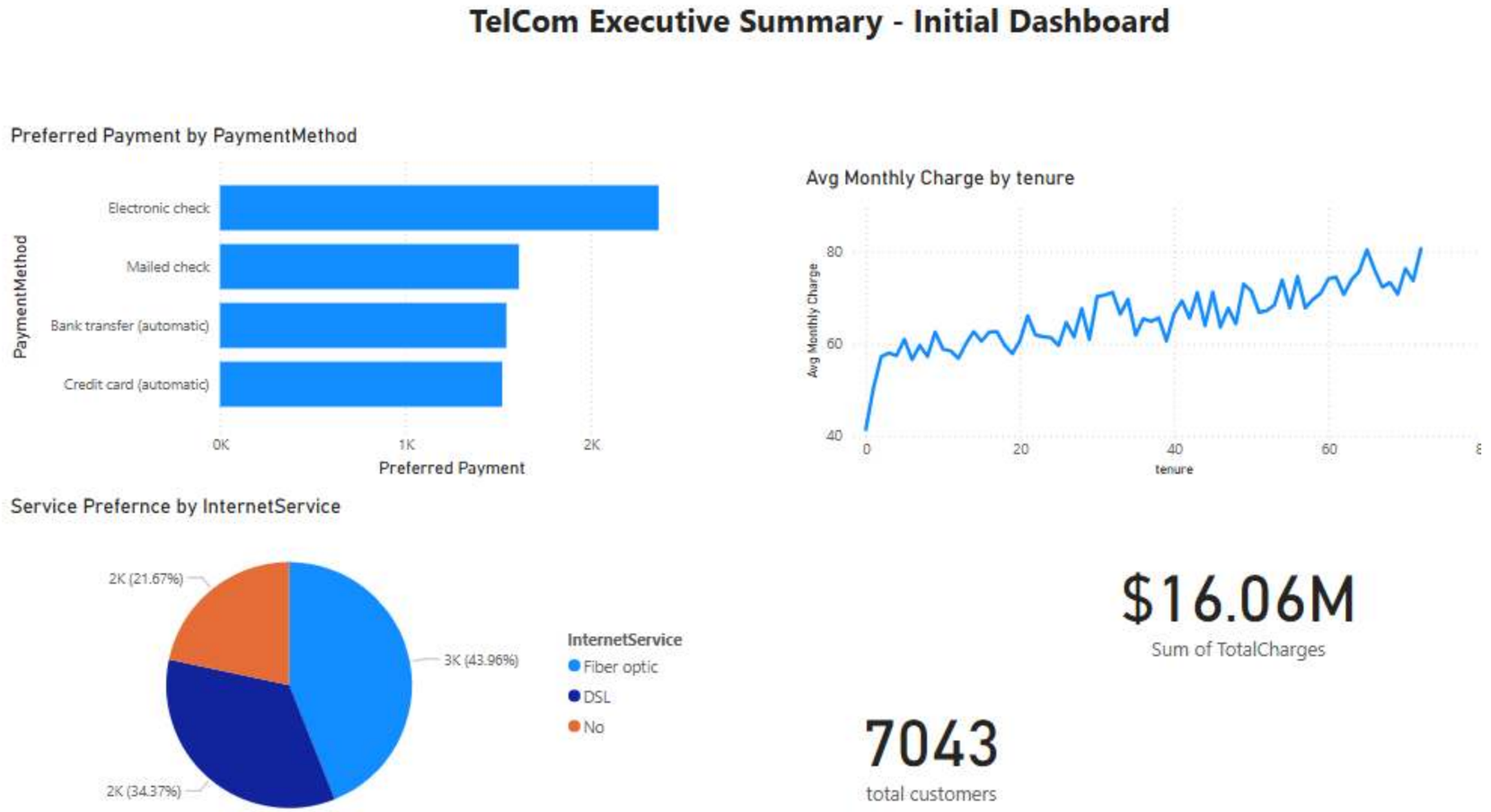
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Data Overview and Preprocessing



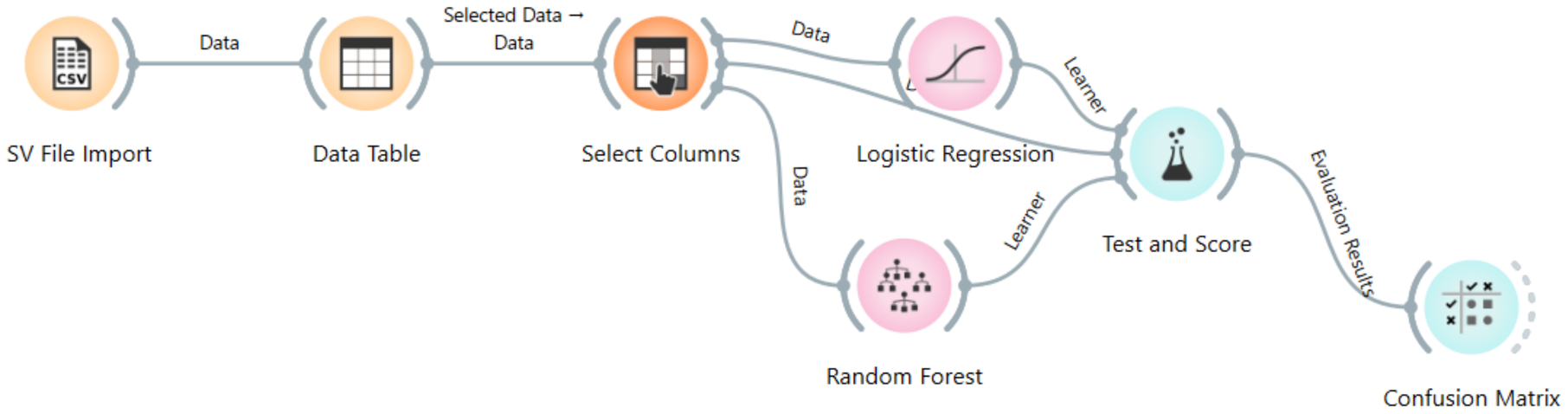
The distributions show no outlier situation of our dataset.

Power BI Dashboard



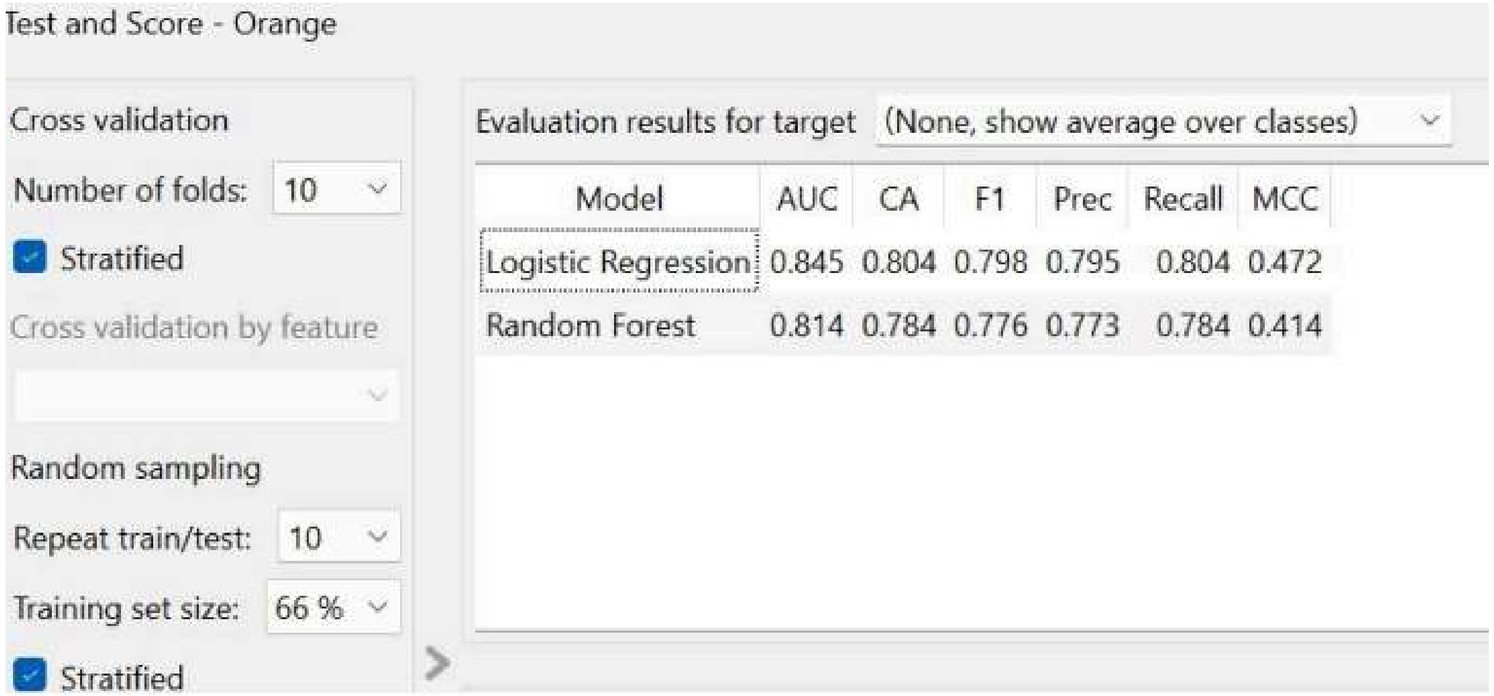
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Our AI Models - Supervised



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Which Model Works Best?

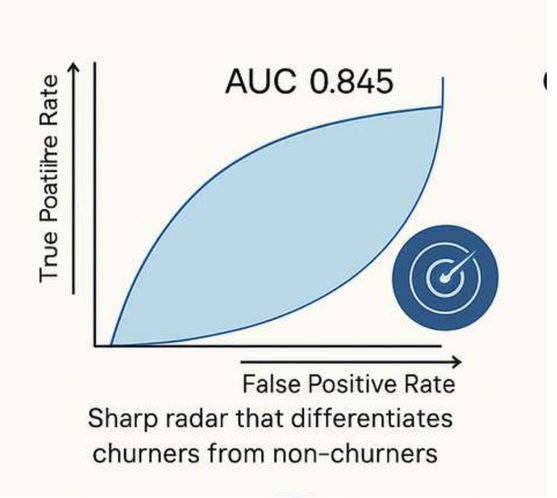


Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.845	0.804	0.798	0.795	0.804	0.472
Random Forest	0.814	0.784	0.776	0.773	0.784	0.414

- Logistic Regression = 80.4% accuracy, 0.845 AUC
- Best balance of precision & recall
- Outperforms Random Forest → cleaner churn signals

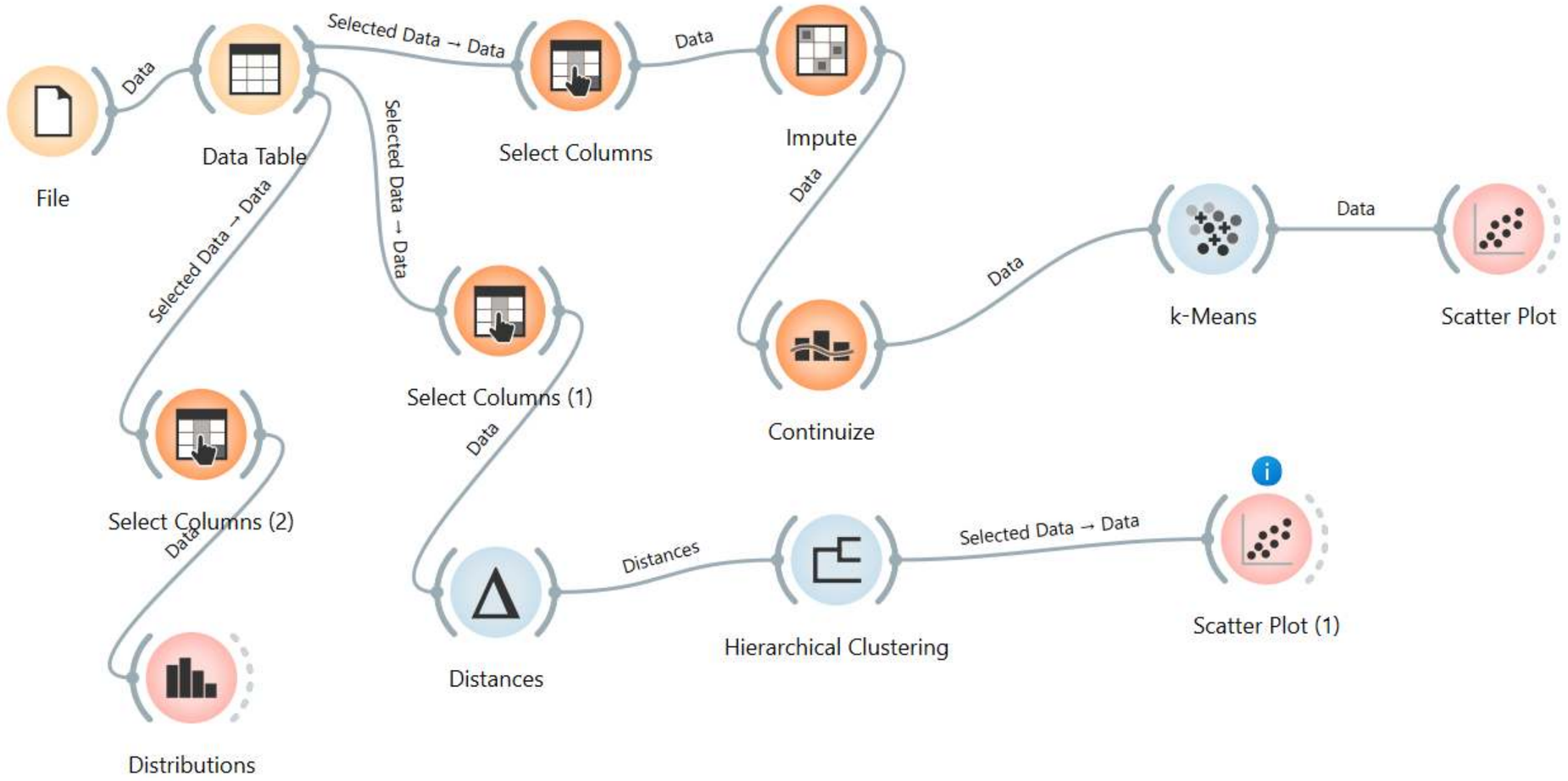
Results and Interpretation

AUC (Area Under Curve): 0.845 (differentiation between churners and non-churners)
CA (Classification Accuracy): 80.4
F1 Score: 0.798
Precision: 79.5%.
Recall: 80.4%
MCC (Matthews Correlation Coefficient): 0.472



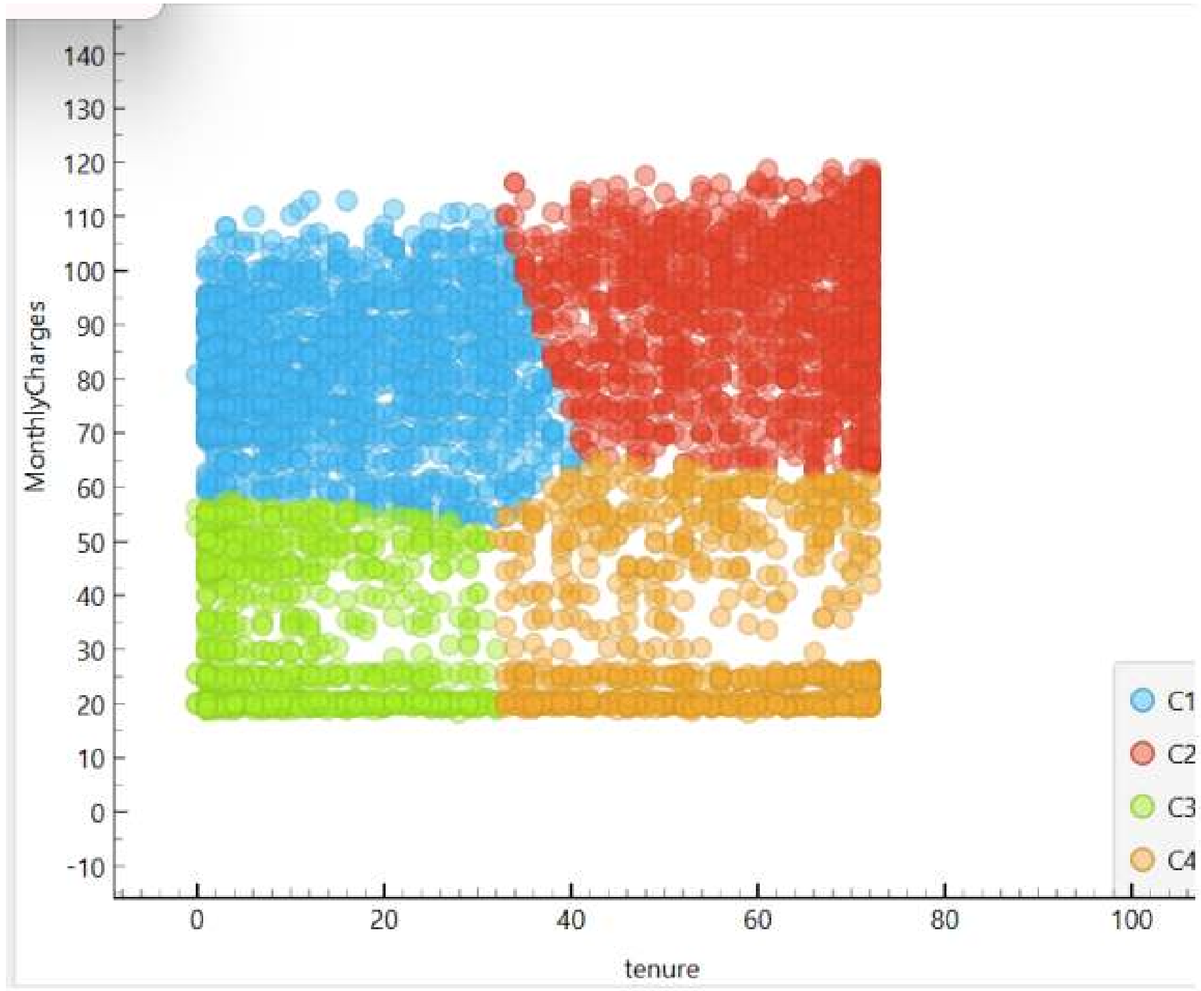
AUC is how sharp the radar is (it identifies churners and non-churners accurately most of the time).
CA is the radar's overall reliability (it's correct in 80% of cases).
Precision is how well the radar avoids false alarms (it doesn't flag loyal customers as churners unnecessarily).
Recall is how well the radar spots all actual churners (it identifies most at-risk customers).
MCC measures how well the radar works overall, considering hits and misses.

Our AI Models - Unsupervised



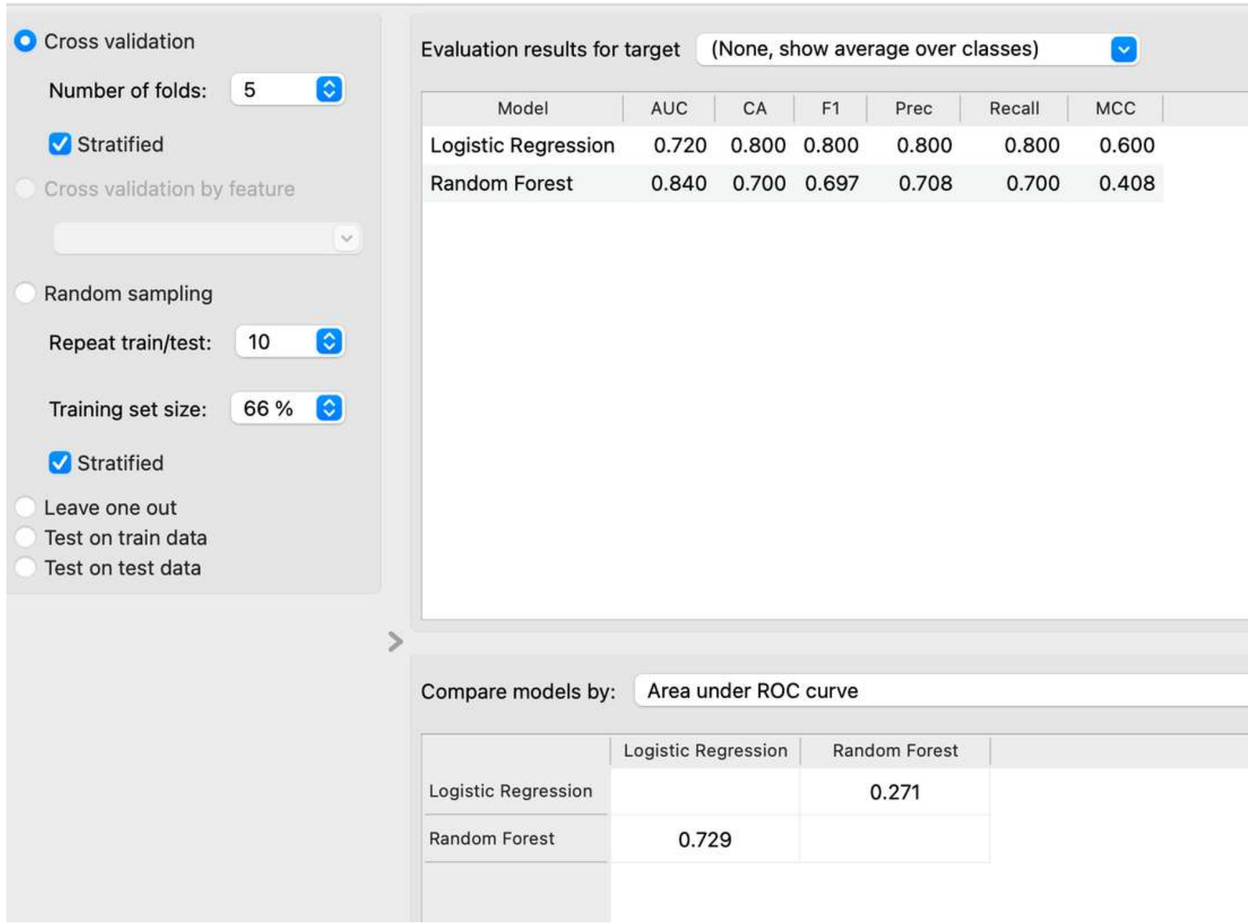
K-Means

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🔵 C1: New, high spend — churn risk.
🔴 C2: Loyal, high value — retain.
🟢 C3: New, low spend — upsell.
🟠 C4: Stable, low value — upgrade.

Predictions



- Random Forest showed a higher AUC of 0.840
- Struggled to make accurate final classifications.
- Relatively lower MCC and F1

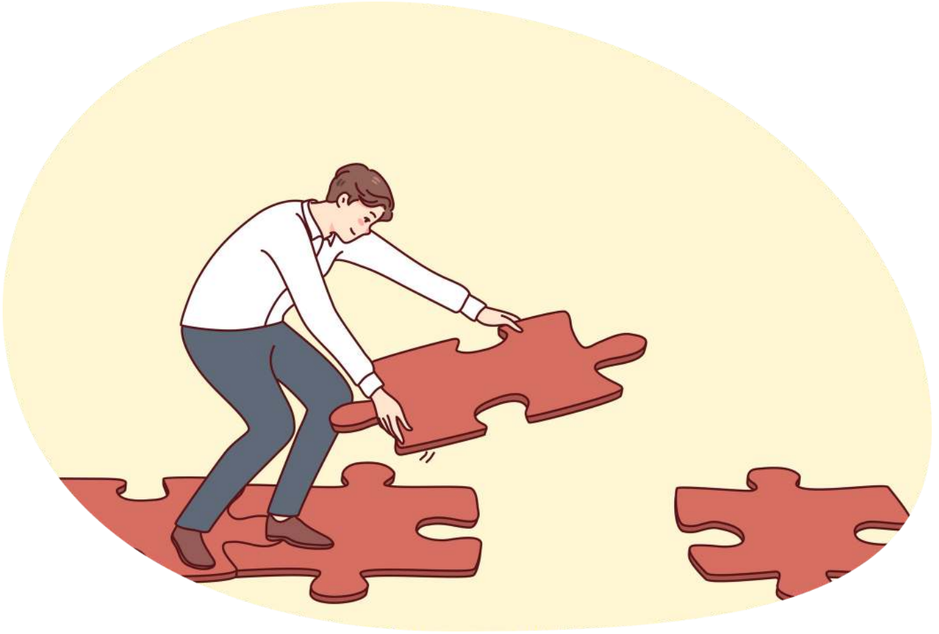
Results and Interpretation

- The model, when tested on the full dataset, can accurately identify which customers are likely to churn.
- It provides a trustworthy churn rate estimation that reflects real business risk.
- Predictions from the 10-sample test can’t be relied on for business decisions, but the 5,000-sample test validates the model's readiness for real-world deployment.
- With this model, the company can now begin to proactively target at-risk customers with retention strategies (discounts, support offers, etc.) based on reliable predictions.

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Key Learnings and Challenges

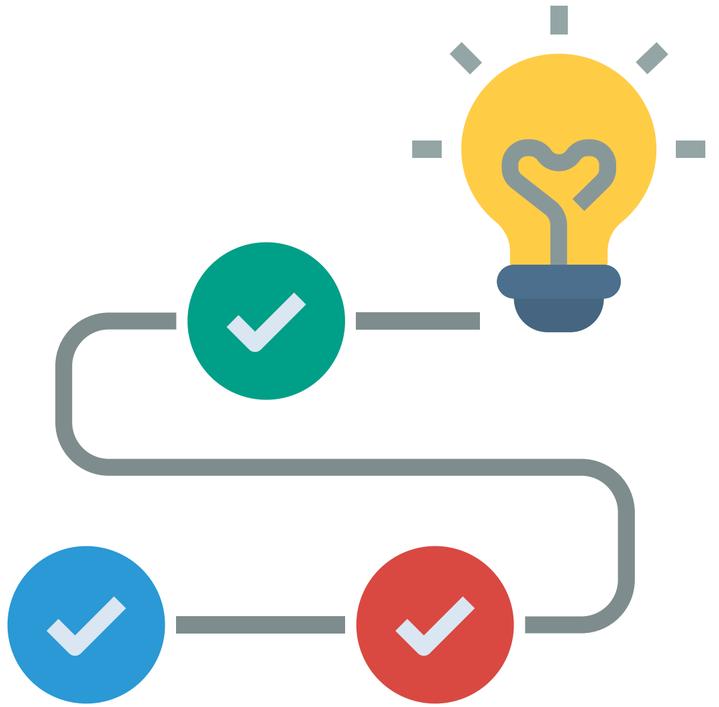
- Had difficulty in outlier detection
- Faced difficulty in calculating centroids
- Trying to understand the difference between precision and accuracy.
- Picking the correct sample.



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Concluding Analysis

- Logistic Regression - AUC of 0.845 and an overall classification accuracy of 80.4%.
- Visualized key customer metrics such as tenure, monthly charges, contract types, and payment methods.
- K-Means clustering - distinguish between customers
- Challenges like precision-recall trade-offs and outlier handling.



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Reference

https://www.kaggle.com/datasets/blastchar/telco-customer-churn

https://www.evidentlyai.com/classification-metrics/explain-roc-curve